Who is who: what matters in injuries in professional team sports

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Abstract

Objective Per-hour injury rates guide rest and match calendars. We measured how they differ between players and with recent load.
Methods Observational study of match records from nine team sports, both sexes. Injury records covered 22 573 players (748 women) in six sports. In football, injuries were substitutions for injury. A heavy week had a full match or more within seven days; a longer-gap week followed an 8-to-13-day gap.
Results Two players' usual injury rates typically differed 1.82-fold. After a full match, the same player's rate typically differed by 5%. Two or more injuries of 28 days or more in three years went with 1.3 to 1.7 times the rate. Heavy weeks went with longer play. Within the same player, playing time rose less. Starters with more recent load stayed on the field longer. These patterns held in every sport we could test. In European men's football, comparing different starters made heavy weeks look 12% safer than longer-gap weeks (0.88, 95% CI 0.82 to 0.94). Within each starter, heavy weeks had 6% more injuries, which could be chance. Injuries confirmed by a recorded absence were a clear 16% more frequent after heavy weeks.
Conclusion Injury rates differ far more between players than with recent league load. Heavy weeks fall mostly to fit regulars, so comparing players mostly shows who carries the load. Compare each player's heavy weeks with the same player's longer-gap weeks, season by season. Medical staff can read repeated injuries of 28 days or more as a risk marker.

What is already known on this topic

- Players differ in injury risk beyond their load and their previous injuries.
- Per-hour injury rates assume that the hours a player records do not depend on the load under study.

What this study adds

- In all six sports with injury records, players differed far more in baseline injury rate than with recent load; in European men's football, heavy weeks looked 12% safer when different starters were compared (0.88), but within each starter, injury-coded substitutions confirmed by a recorded absence were 16% more frequent after a heavy week (1.16, 95% CI 1.03 to 1.29).

How this study might affect research, practice or policy

- Researchers and injury surveillance programmes should compare each player's heavy weeks (a full match or more) with the same player's longer-gap weeks (8 to 13 days between matches) in the same season, count injuries confirmed by a recorded absence and, where they infer injury from team sheets, should not read a missed start in the next match as injury.
- Club medical staff can read two or more injuries of 28 days or more in three years as a marker of higher risk (in football, 4.4 against 3.1 injury-coded substitutions per 100 starts), which confirms a known risk factor: players with this history came through most matches uninjured, but were injured more often.

Introduction

Injury rates guide which players medical teams rest and how many matches players play.[1, 2] Both uses assume that recent load, meaning last week's play, drives injury risk. Yet some players are injured more often than others. A previous injury, for example, is a strong risk factor for the next one.[3] Suppose the players who play the most are also those injured least. Then injury rates that compare players will make a heavy week, a full match or more in the previous week, look safe.

In football, injuries are counted per 1000 hours of play.[4–6] Studies of fixture congestion and recent load use such rates.[7–12] All of them assume that the hours a player records stay the same, whatever the recent load. But players who played more in the previous week play more minutes in the next match.[13, 14] We call this rise the workload gradient.

A per-hour rate divides injuries by the minutes played. The number we divide by is called the denominator. After a heavy week the minutes rise, so the denominator rises too. The rise could have two causes. A heavy week might change how long a player plays. Or the players with a heavy week might be the regular starters, those the coach picks to start every week and keeps on the field. We call this choice team selection. Which cause holds decides whether a per-hour rate compares loads or compares players.

Earlier work shows that players differ in injury risk beyond their load and their previous injuries.[15] In basketball, coaches rest players when their injury risk rises.[16] In medicine, comparing groups of people who were picked in some way can mislead.[17] Some injury research has corrected its rates by counting only the players fit to play.[18] Rates per appearance and per hour can also give different answers.[19–21] But to our knowledge, only the first author's nine-sport preprint[22] has measured how much of the recorded playing time comes from team selection. Others have raised the question.[1, 23]

In nine team sports, we asked how much injury rates differ between players and with recent load, and why comparing players can mislead. In 99 first-tier men's football competitions, we asked whether coaches keep starters on after a heavy week.

Methods

Design and data

This observational study used public match records: it studied matches as they were played. The records show each team's line-up and the minute of each substitution. The study follows the STROBE reporting guideline.[24] The records of men's football come from Transfermarkt. We call it the provider. Table 1 lists the nine sports and names the leagues. We asked the same questions of 37 more leagues in seven sports, rugby league and cricket among them (Supplementary Section S9). In 2020, most football leagues moved from three substitutions per match to five.[25] A league sample is one league's seasons under one substitution limit (Table 1; Supplementary Section S2 and Table S1).

Recent match load

Recent match load, or recent load for short, is the league minutes a player played in the previous seven days. It is counted in full matches of 90 minutes. A player with a full match or more was loaded: a heavy week. A player with some minutes, but less than a full match, played part of a match. A player with no minutes was rested. The player was rested by the calendar when the club had no league match that week. The player was absent when they missed the club's previous match. A longer-gap week is a week rested by the calendar that came 8 to 13 days after the club's previous league match. In a longer-gap week, the fixture list set the rest, rather than the coach or an injury. So it makes a fair comparison for a heavy week.

Injury records

A substitution is injury-coded when the match report gives injury as its reason. We used only league-seasons that allowed five substitutions, and in which at least 90 per cent of substitutions have a reason. We say these league-seasons have near-complete reasons. The provider's injury histories show a recorded absence within a week after 39% of injury-coded substitutions, and after 4% of tactical ones. An absence is a sign of real injury, so we also counted only the injury-coded substitutions that a recorded absence followed. We call this second count the narrower outcome. Every football injury result is given on both outcomes.

A statistical model gave each player their own usual injury rate, which we call the baseline rate. It takes in everything that stays the same about a player, such as position, usual load and how the club records injuries (Supplementary Section S11). The same model also measured how a player's own rate differed with each extra full match of recent load. We ran it in each of 11 player groups (Table 1). In five sports, the records give the length of each absence. In those sports, we compared players by their injury history (Supplementary Section S12).

The injury comparison

We counted injury-coded substitutions in three ways: per appearance, per 1000 hours played, and, for starters, per 1000 hours played as a starter. We call these hours recorded hours and starter hours. We compared heavy weeks with rested weeks, or with longer-gap weeks only, in two ways. Across players, we compared all players' weeks in each league-season, and then combined the league-seasons (Mantel–Haenszel). Within each starter, we compared each starter's heavy weeks with the same starter's own rested or longer-gap weeks (conditional Poisson regression). We did this in the same league-season, and also with all of a starter's seasons pooled. One more model, within each starter, compared injury rates by the days since the club's previous match in any competition (Supplementary Table S21). A computer simulation tested how well each comparison finds a known answer (Supplementary Section S6).

The workload gradient

The workload gradient is how much a player's recorded minutes rise with recent load. A gradient of 0.03 means about three per cent more minutes per extra full match. We split it into two parts: one comes from who starts, the other from starters and substitutes playing longer.[26] For starters, the within-player slope is the gradient measured within each starter. It shows whether the same starter played longer after their own heavy weeks (Supplementary Section S3).

When starters came off

We modelled the chance that a starter was taken off in each part of the match, by their recent load. We ran this model in the eight European men's leagues and in a worldwide survey of 162 samples from 99 first-tier men's competitions (Table 1; Supplementary Figure S1 and Table S3). In the eight leagues, a second version compared each starter's heavy weeks with the same starter's other weeks.

Statistical analysis and pre-set criteria

The statistics follow the CHAMP statement.[27] We set two criteria before testing them (Supplementary Section S4). First, from sample to sample, the within-player slope would vary at most half as much as the rise across starters. Second, the more firmly a league keeps loaded starters on, the more of the starters' rise would come from differences between starters. Every other analysis, the injury comparison among them, is exploratory: it finds leads to check. But we chose the comparison the design favours, each starter against themselves in the same season, after seeing the results of similar comparisons. A confirmation on the next complete seasons is registered, with a plan written down in advance. Each result is reported on its own, as if it were the only comparison. We wrote the analysis scripts with an AI assistant and checked them against hand-worked tests.

How to read the results

Most results are ratios of two injury rates. A ratio of 1.16 means a rate 16% higher than the rate it is compared with, and 0.88 means 12% lower. Each ratio comes with a range, its 95% confidence interval (95% CI). The true value is likely to be in this range. When the whole range is above one, or below one, the difference is clear. When the range includes one, there is no clear difference. For heavy weeks, we call a clear rise riskier and a clear fall safer.

Equity, diversity, and inclusion statement

The league samples include women's football leagues, and women's basketball and cricket. The worldwide survey covers men's leagues only. Women's injury records come from one league, the WNBA: 748 of the 22 573 players with injury records. The two authors work at one European university, in statistics and medicine.

Patient and public involvement

Patients or the public were not involved in the design, conduct, reporting or dissemination plans of this research.

Results

Injury rates differed far more between players than with recent load

In 10 of the 11 player groups, players differed from each other far more than the same player's rate differed after a full match (Figure 1A; Supplementary Section S11). Across these 10 groups, two players picked at random typically had baseline rates 1.82 times apart. After a full match, the same player's rate typically differed by 5%, up or down. That 5% is for one whole extra match, and most weeks a player's load changes by less. So through load alone, two weeks of the same player typically had rates only 1.02 times apart. In football, the narrower outcome gave the same picture (Figure 1A; Supplementary Section S11).

But a single start is mostly chance. In football, a player's baseline rate accounted for only about 2% of the variation in whether one start ended in an injury-coded substitution.

Injury rates also differed with a player's injury history. In every sport with injury histories, some players had two or more injuries of 28 days or more in the previous three years. They had 1.3 to 1.7 times the injury rate of the others (Figure 1B; Supplementary Section S12).

Heavy weeks looked safer across starting players than within each one

About one start in 31 ended in an injury-coded substitution (0.0323 per start). Among outfield players, loaded players had 26% more injury-coded substitutions per appearance than rested players (1.26, 95% CI 1.19 to 1.33; Figure 1C and Figure 2). But loaded players also played longer. So per hour played, they had 9% fewer (0.91, 0.86 to 0.97).

Next we compared starters only, per starter hour, against longer-gap weeks. Across starters, heavy weeks looked 12% safer (0.88, 95% CI 0.82 to 0.94). With the narrower outcome, they showed no clear difference (0.94, 0.85 to 1.05). But within the same starter and season, heavy weeks had a 6% higher rate, with no clear difference (1.06, 0.99 to 1.13). With the narrower outcome, heavy weeks were riskier: the rate was 16% higher (1.16, 1.03 to 1.29).

Within the same starter, we also compared gaps since the club's previous match: two to three days against six to seven. After the short gap, the narrower outcome was 40% more frequent (1.40, 1.23 to 1.59; Supplementary Table S21). With injury-coded substitutions, the short gap showed no clear difference (1.05).

Who starts accounts for most of the workload gradient

Who starts accounted for 73% to 92% of the workload gradient in all 18 samples of football and rugby union (Supplementary Figure S3A). Within each starter, recorded minutes changed little (Supplementary Figure S3B).

Across starters, those with a full match in the previous week stayed on longer

Starters who had played a full match in the previous week were taken off less often. This was clear in all 162 samples of the worldwide survey and in all 15 of the other leagues that recorded substitution

Table 1: Which data answer which question. The nine sports are football, rugby union, Australian rules, basketball, ice hockey, American football, rugby league, baseball and cricket. The eight European men's leagues are those of England, France, Germany, Italy, the Netherlands, Portugal, Spain and Turkey. League-seasons, players and sizes are totals over the samples. In the rows on players against recent load, the league samples are the 11 player groups, one or more to a row. A player in two samples of one question counts twice. The exception is a source that gives each player one identifier across both samples: the NBA with the G League, and MLB's batters with its pitchers. The worldwide survey, the injury comparison and the next start count each player once. Each of the worldwide survey's 162 samples has its own result. Its pooled results combine the 140 samples played with three or five substitutions allowed (Supplementary Table S24). The next start asks how often a starter taken off early started the club's next match (Supplementary Section S3). Outside football, an injury is each league's own record, counted once per absence (Supplementary Section S12). Median age is the range of the samples' median ages at appearance; a dash marks rows without one (Supplementary Table S2). The injury comparison's row includes goalkeepers, whom Figure 1C and Figure 2 leave out (19 231 starts).

Question	Sport and sex	League samples	League-seasons	Players	Of them, women	Median age, years	Size	Unit of size
Players against recent load (Figure 1A) and injury history (Figure 1B)	Football, men: England, France, Germany, Italy and Spain, as one group	1	27	4591	0	–	6710	injury-coded substitutions, outfield starters
	Australian rules, men	1	9	1253	0	–	1824	absences through injury
	Basketball, men and women	4	62	4252	748	–	3180	absences through injury; injured-list placements in Japan
	Ice hockey, men	2	13	1932	0	–	1946	absences through injury
	Baseball, men, batters and pitchers	2	32	3965	0	–	4743	injured-list placements
	American football, men	1	13	6580	0	–	4613	absences through injury
Who starts (Figure S3A)	Football, 10 men's and 7 women's; rugby union, men	18	74	18 419	2104	24.5–28.7	685 148	appearances
Within the same player (Figure S3B)	Football, men and women; rugby union, men; Australian rules, women	25	57	18 673	3265	24.4–28.5	14 963	players with two or more starts
Kept on, worldwide survey	Football, men's first tiers in 96 of 99 competitions, the eight European men's leagues fitted apart	140 of 162	538	72 208	0	–	2 089 384	starter appearances
The injury comparison (Figures 1C and 2) and one start in 31	Football, men, the same five leagues	5	27	5942	0	25.4–27.4	296 029	appearances, goalkeepers included, with 7158 injury-coded substitutions
The next start	Football, men, 14 leagues, five substitutions	14	82	13 202	0	24.8–28.7	490 125	starts followed by another club match

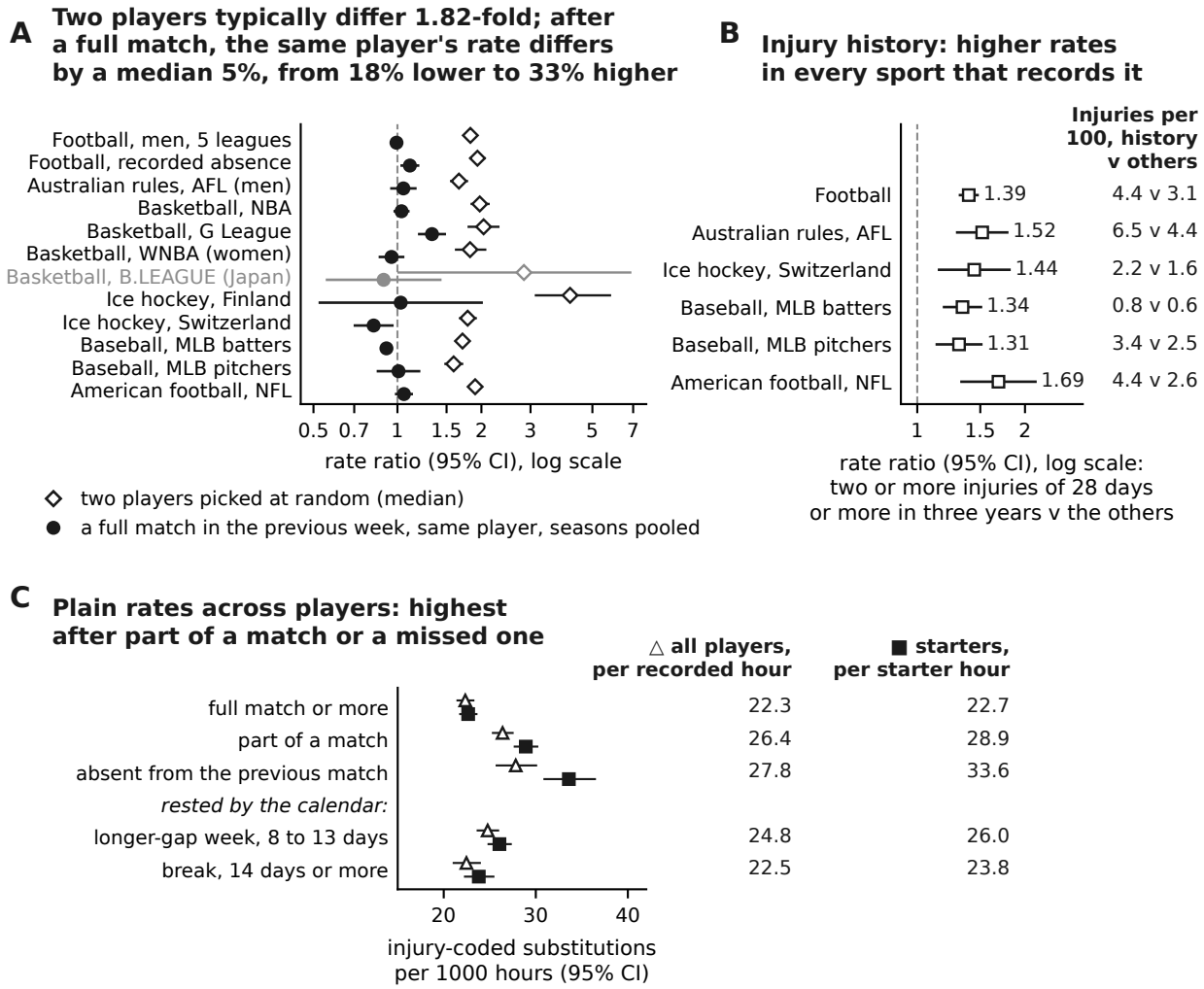


Figure 1: Injury rates differed far more between players than with recent load. (A) For each player group, the open diamond shows how far apart two players picked at random typically are. This is the median rate ratio: the higher rate over the lower. The filled dot shows how the same player's rate differed after a full match in the previous week, with seasons pooled. Lines show 95% CIs (Supplementary Table S29). On the log scale, doubling and halving sit the same distance from one. Football counts outfield starts only, once for injury-coded substitutions and once for those a recorded absence followed. The G League leaves out absences that were likely moves to the NBA. Japan's B.LEAGUE, drawn lighter, had only 27 injuries. Its ranges are so wide that the two measures cannot be told apart there, even though its markers sit far apart. Finland's ice hockey league, the Liiga, has the highest ratio between players. Its teams record injuries in different ways, and allowing for each team, its ratio is 2.91 (Supplementary Section S12). Outside football, some injury lists record rest, or a move to another team, as an injury. That would move both kinds of marker to the right. (B) Players with two or more injuries of 28 days or more in the previous three years, against the others. Each row gives the rate ratio: per start in football, and per unit of time played elsewhere. Beside each ratio are the two groups' injuries per 100 starts or appearances (Supplementary Table S30). (C) Plain injury rates, before any comparison within players. They are injury-coded substitutions per 1000 hours among outfield players in the 27 league-seasons with near-complete reasons, with 95% CIs. Triangles count all players per hour played, and squares count starters per starter hour. Players absent from the previous match are a separate kind of rest from the weeks rested by the calendar.

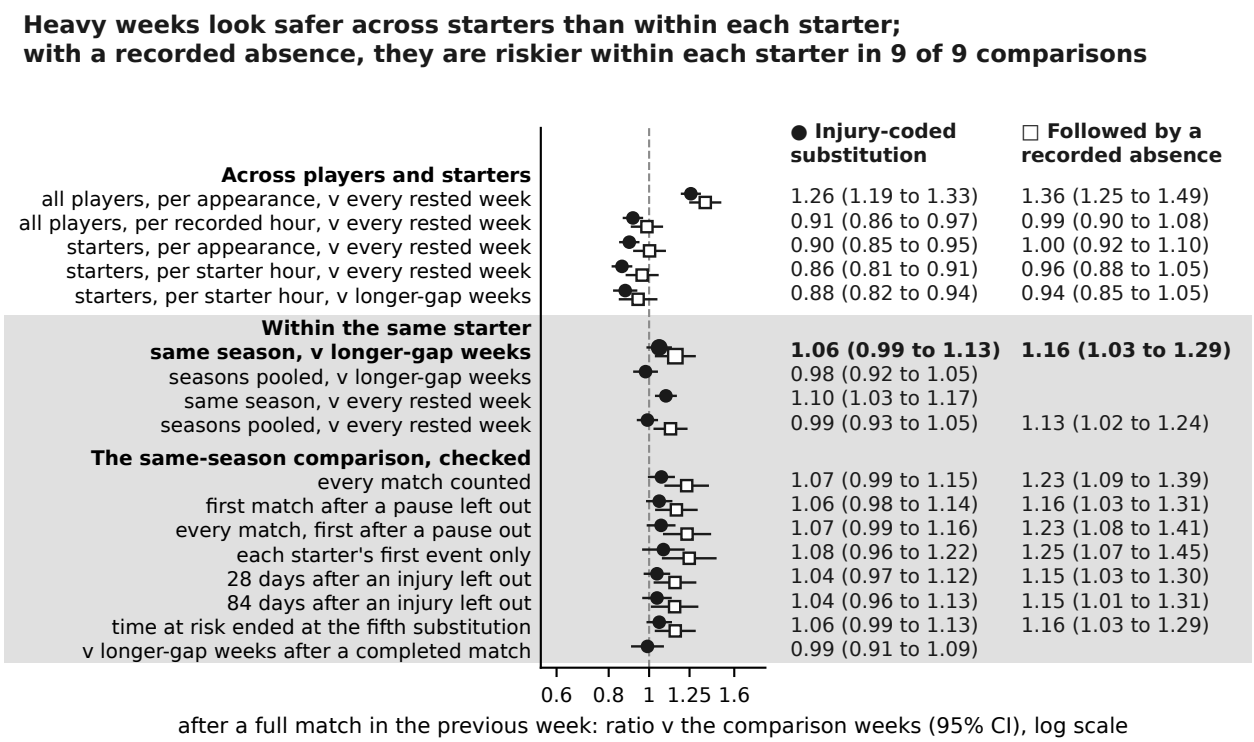


Figure 2: Heavy weeks looked safer across starting players than within each one. Outfield players in the 27 league-seasons with near-complete reasons. Each row gives the ratio after a full match or more in the previous week, with 95% CIs. Filled circles count injury-coded substitutions, and open squares count those a recorded absence followed. Heavy weeks look safer where the whole interval lies below one, and riskier where it lies above one. Unshaded rows compare all players, or all starters, with each other. Each compares weeks within a league-season and then combines the league-seasons (Mantel–Haenszel). Their rates are per appearance or per hour, against every rested week or against longer-gap weeks. A longer-gap week was rested by the calendar, 8 to 13 days after the club's previous league match. Shaded rows compare each starter's own weeks (conditional Poisson regression), within the same league-season or with seasons pooled. The row in bold is the comparison the design favours, because it sets each starter against themselves in the same season. The checks repeat it. One counts every match the provider records. Another leaves out each club's first league match after a pause in the league. Most pauses are international breaks, when international players played for their countries. Another keeps only each starter's first injury-coded substitution. Two more leave out the 28 or 84 days after an injury-coded substitution, when a player is coming back. The time at risk is the minutes in which an injury can still end in a substitution. Another check ends it at the team's fifth substitution. The last uses only the longer-gap weeks after a completed match, since a player who completed the previous match did not come off early with a knock. Three rows have no square: those comparisons were run on injury-coded substitutions only. Supplementary Table S20 gives the counts.

times (Supplementary Table S26). It was also clear within the same player. In the eight European men's leagues, the same starter's odds of being taken off fell by 10% when three substitutions were allowed (0.90). They fell by 14% when five were allowed (0.86; Supplementary Table S17). Part of this result is about roles. Some starters usually play the whole match. They had a full match last week, and they stay on this week too (Supplementary Section S6).

A summary of our key findings is presented in Figure 3.

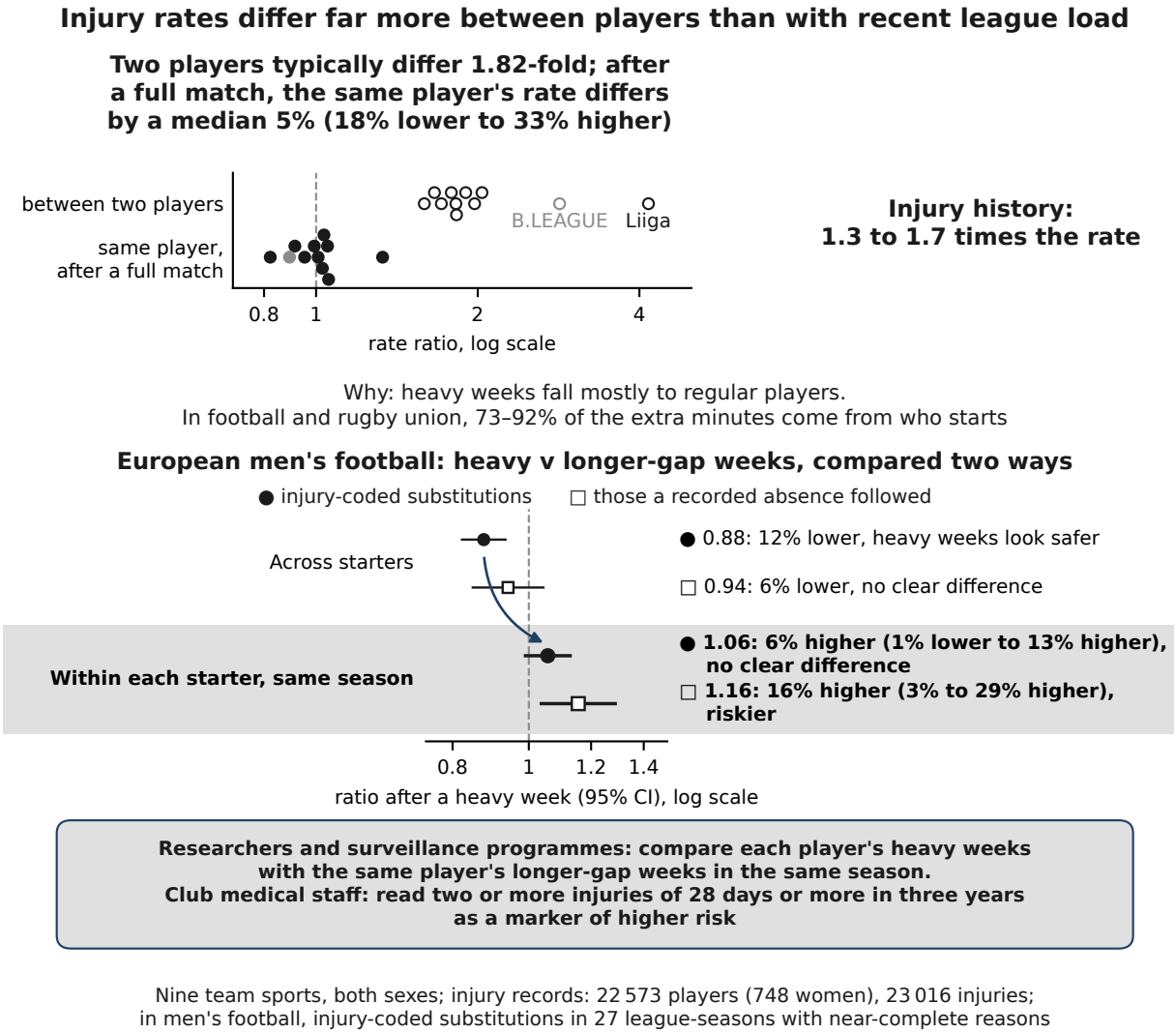


Figure 3: Central Figure: injury rates differ far more between players than with recent league load, and how to compare like with like. In the top row, each dot is one of the 11 player groups, moved apart where dots would overlap. An open dot gives the median rate ratio between two players picked at random. A filled dot gives the same player's rate ratio after a full match in the previous week, seasons pooled (Supplementary Table S29). The open dot at the far right is Finland's Liiga, whose teams record injuries in different ways. Japan's B.LEAGUE, drawn lighter, had too few injuries for its two measures to be told apart. The injury history's ratios span the six groups that record it. In the second row the symbols change meaning. A filled mark shows outfield starters' injury-coded substitutions per starter hour, and an open mark shows those a recorded absence followed. Both are against longer-gap weeks: across starters, and within each starter (shaded) in the same league-season. The words beside each interval give its reading: safer or riskier when it lies wholly below or above one, no clear difference when it holds one. The arrow runs from the standard comparison, across starters, to the comparison within each starter. A heavy week had a full league match or more in the previous seven days. A longer-gap week followed a gap of 8 to 13 days. The box gives this study's own advice.

Discussion

Injury rates differed far more between players than with recent load. Heavy weeks fall mostly to regular starters, the players fit enough to keep their place. So in European men's football, heavy weeks looked safer across starters than within each one.

Per-hour rates divide by who starts

In football and rugby union, most of the extra minutes after a heavy week came from who starts. So a per-hour rate compares different players, as well as different loads.[28] For a regular starter, a heavy week is the usual week. In the eight European men's leagues, 59% to 68% of a club's matches came within seven days of its last one. Rotation players and substitutes play a full match less often. So most heavy weeks belong to regular starters. Workplace health research calls this the healthy worker effect: the workers who do most of a job are those fit enough to stay in it.[29–31]

Two recent papers in this journal discuss counting per appearance.[32, 33] That count does not remove this selection among starters (Figure 2). Measuring how much minutes rise with load, as our earlier preprint did,[13] cannot show who carries the load either.

How comparing players misleads

- **Heavy weeks look safe.** Comparing starters with each other gives the heavy week the credit for the regular starters' lower risk. This matches a healthy worker effect.[34]
- **The denominator can flip the answer.** Loaded players had more injury-coded substitutions per appearance than rested players, but fewer per hour played.
- **Comparing players by their load misses those at higher risk,** such as players with repeated injuries of 28 days or more.

Other factors in resting a player

Besides recent load, a decision to rest a player can weigh:

- previous injury, a strong risk factor;[3]
- recovery time: among rested starters, rates were lowest after breaks of 14 days or more;
- every match played, and the travel between them;[10]
- training load, which match records never show;[11]
- squad depth and rotation.[8, 35]

Implications for research

First, the comparison the design favours sets each player against themselves in the same season. Report it beside the standard comparison across players. It removes stable differences between players, but selection that changes from week to week remains.

Second, count injuries confirmed by a recorded absence, and count load over every match played (Figure 2).

Third, researchers who infer injury from team sheets should not read a missed start as injury (Supplementary Section S3).[36]

Clinical implications

First, two or more injuries of 28 days or more in three years mark a group at higher risk, not the players who will be injured.[37] In football, players with this history made 26% of outfield starts but had 33% of injury-coded substitutions: 4.4 against 3.1 per 100 starts (Supplementary Section S12). The history confirms a known risk factor and should not by itself change selection.

Second, a small difference with recent load does not show that congestion is safe. Within the same starter, the narrower outcome was 16% more frequent after a full match. This result comes from one sport and from an exploratory comparison.

Limitations

The most important limitation is how recent match load was measured.[38] The main analysis counts league matches only. Matches in the international breaks are never counted. So an international player's load is understated in exactly the weeks that congestion research studies (Supplementary Table S12).

Second, the records are public match data, not surveillance data kept by club doctors.[39, 40] The injury reason is a label from the match report, not a diagnosis. It also covers players taken off as a precaution. The staff whose choices this study measures make that call (Supplementary Section S3). A recorded absence followed injury-coded substitutions a little more often after a heavy week (41%, against 34% to 39% after other weeks). So the narrower outcome keeps a few more of the heavy weeks' injuries, and may make heavy weeks look slightly riskier. An absence can also be a staff decision (Supplementary Section S3). No video audit of the study seasons was feasible.

Third, the design is observational: it records what happened and changes nothing.[41] Selection from week to week can still bias the result. For example, a player who came off early with a knock may carry more risk into the next start. A computer simulation measured that bias, and the real data showed no sign of it (Supplementary Section S6). The design also limits how small a change it can find. Recorded minutes stop at the full match, so the smallest within-player slope the design could detect was 0.006 to 0.024 (Supplementary Section S6).

Fourth, we saved the two pre-set criteria, with a date, to the project's own files, not to a public protocol. A public protocol is a plan published in advance. We saved both after our earlier preprint had used the same records. The first passed narrowly and the second failed. On the first, the within-player slope varied 0.48 times as much as the rise across starters, against a limit of one half. The second failed because its link was too uncertain to tell either way (Supplementary Section S4).

Finally, the samples differ in sport, sex, era and size. The only women's injury record is one basketball league. The injury comparison comes from European men's football (Supplementary Section S9).

Conclusion

Injury rates differ far more between players than with recent league load. Heavy weeks fall mostly to regular players, who are fit enough to keep their place. So injury rates that compare players mostly show who carries the load. Within each starter, injury-coded substitutions confirmed by a recorded absence were more frequent after a full match. They were also more frequent after a gap of two to three days. A registered confirmation on the next seasons will test the heavy-week comparisons. Compare each player's heavy weeks with the same player's longer-gap weeks in the same season. Read repeated injuries of 28 days or more as a marker of risk.

Declarations

Contributors GPR designed the study, collected and analysed the data and wrote the manuscript. NM shaped the methods and the clinical framing of the earlier preprint that this manuscript extends. Both authors approved the final version. GPR is the guarantor. Anthropic's Claude, a generative

AI assistant, was used under GPR's direction to write and check analysis code, search the literature and edit the text. It was used because the first author could not otherwise build and check analyses of this size. The study design, the claims and their interpretation are the authors'. Every reported quantity is computed from the committed data by the archived code and was verified by GPR. Both authors take full responsibility for the content.

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Competing interests None declared.

Prior dissemination This manuscript is posted on arXiv as the next version of our earlier preprint.[13] That preprint raised the question this manuscript answers. Some of this manuscript's results were first posted in the first author's nine-sport preprint.[22]

Ethics approval The study reused only public, pre-existing match records. There was no recruitment, no contact with any individual, no intervention and no access to medical records or clinical files, so informed consent did not apply. The records name professional athletes, and some reasons for substitution are injuries. They were therefore treated as special-category personal data despite being public. Processing was for scientific research under the research provisions of the General Data Protection Regulation. Everything published, every table, figure and deposited file, is aggregated to whole league samples, so none of it describes a single player. Ethical approval was not sought. These published records were treated as archival data, which Trinity College Dublin's research ethics guidance excepts from approval.[42] The same guidance grants no approval retrospectively.

Data availability statement Data are available in a public, open access repository. The aggregated results behind every table and figure, and the analysis scripts, are in the archived record at DOI RESERVE-BEFORE-SUBMISSION. No player-level record is shared. The match records the scripts read identify players, so they are not redistributed. Their reconstruction from the public providers, each source's terms of use as read and any permission held are documented in the archive. Statistics from NBA.com are credited to NBA.com, as its terms ask.

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